



Java Learning Center

No.1 In Java Training & placement

DSA

Module 1

Introduction to DSA

Author

Srinivas Dande





Java Learning Center

No.1 In Java Training & placement

1. Data types

- ♦ Data type represents:
 - Type of data you want to use
 - Amount of memory allocation required for your data.

There are two types of data types

- 1) Primitive Data Types
- 2) User Defined Data Types

Primitive Data Types

- ♦ There are 8 primitive types available in Java.
- ♦ 8 keywords are defined to represent 8 primitive data types.
- ♦ Following are primitive data types:

boolean	byte	char	short
int	long	float	double

User-Defined Data Types

- ♦ There are four types of User Defined Data types:
 - Class type
 - Interface type
 - Enum type (From JAVA 5)
 - Annotation type (From JAVA 5)

2. Variables

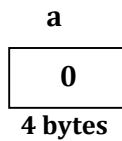
- ♦ Variable is the container which holds user data.
- ♦ Memory will be allocated for the variable while executing the program.
- ♦ Value of the variable can be changed any number of times during the program execution.

Syntax:

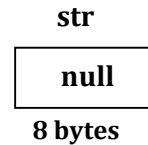
```
<Data type> <varName>;  
<Data type> <varName> = <value>;
```

Ex:

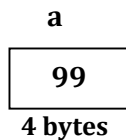
`int a ;`



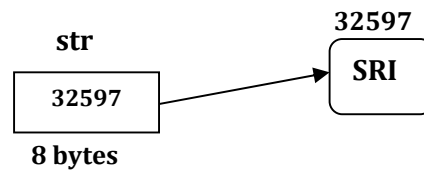
`String str;`



`int a = 99;`



`String str="SRI";`



Types of Variables

There are two types of variables based on data type used to declare the variable.

- 1) Primitive Variables
- 2) Reference Variables

2.1 Primitive Variables

- ♦ Variables declared with primitive data types are called as primitive variables.

Ex:

`int a;`
`int b = 99;`

2.2 Reference Variables

- ♦ Variables declared with user defined data types are called as reference variables.

Ex:

`String str1;`
`String str2 = "JLC";`



3. Data Structures

- ♦ Data Structure is a special way for storing and organizing data so that it can be used efficiently.
- ♦ Following are Types of Data Structures
 - Arrays
 - Linked Lists
 - Stacks
 - Queues
 - Trees
 - Graphsetc
- ♦ We can divide the Data Structures into two Types depending on the organization of the elements
 1. Linear Data Structures
 2. Non - Linear Data Structures

3.1 Linear Data Structures

- ♦ Elements are accessed in a sequential order but it is not compulsory to store all elements sequentially.

Ex:

Linked Lists

Stacks

Queues

3.2 Non - Linear Data Structures

- ♦ Elements of this data structure are stored/accessed in a non-linear order.

Ex:

Trees

Graphs



4. Abstract Data Types

- ♦ Define Data Type by Combining the data structures with their operations which is called as Abstract Data Type (ADT).
- ♦ An Abstract Data Type(ADT) is an abstraction of a Data Structure that provides only the interface to which the Data Structure must adhere.
- ♦ The interface does not give any specific details about how something should be implemented or in what programming language.
- ♦ ADT consists of two parts:
 1. Declaration of data
 2. Declaration of operations
- ♦ Following are Commonly used ADTs:
 - Linked Lists
 - Stacks
 - Queues
 - Priority Queues
 - Binary Trees
 - Dictionaries
 - Disjoint Sets
 - Hash Tables
 - Graphs
 - etc
- ♦ Different kinds of ADTs are suited to different kinds of applications, and some are highly specialized to specific tasks.
- ♦ By the end of this Course, we will go through many of them and you will be in a position to relate the data structures to the kind of problems they solve.

5. Exploring Algorithms

- ♦ An algorithm is the step-by-step unambiguous instructions to solve a given problem.
- ♦ There are two main criteria for judging the algorithms:
 - a) Correctness
 - b) Efficiency

a) Correctness

- ♦ Does the algorithm give solution to the problem

b) Efficiency

- ♦ How much resources (in terms of memory and time) does it take to execute

Examples:

Ex1: Algorithm for Adding Two Numbers

```
/*  
* @Author : Srinivas Dande  
* @Company: Java Learning Center  
* */
```

Step 1: Start

Step 2: Declare 3 Variables a , b and sum.

Step 3: Read values for a and b

Step 4: Add a and b and assign the result to a variable sum.

Step 5: Display sum

Step 6: Stop



Ex2: Algorithm for Adding Two Numbers

```
/*  
* @Author : Srinivas Dande  
* @Company: Java Learning Center  
* */
```

Step 1: Start

Step 2: Declare 3 Variables a , b and sum.

```
int a;  
int b;  
int sum;
```

Step 3: Read values for a and b

```
a = 10;  
b = 20;
```

Step 4: Add a and b and assign the result to a variable sum.

```
sum = a+b;
```

Step 5: Display sum

```
print(sum);
```

Step 6: Stop



Ex3: Algorithm for Printing Numbers from 1 to 10

```
/*  
* @Author : Srinivas Dande  
* @Company: Java Learning Center  
* */
```

Step 1: Start

Step 2: Declare and Initialize the Variable i to 1

```
int i=1;
```

Step 3: Display i

```
print(i);
```

Step 4: Increment i by 1

```
i = i + 1;
```

Step 5: Check if the value of i is less than or equal to 10.

```
IF i<=10 THEN
```

```
    GOTO Step 3
```

```
otherwise
```

```
    GOTO Step 6
```

Step 6: Stop.

6. Algorithm Analysis

- ♦ Algorithm analysis helps us to determine which algorithm is most efficient in terms of time and space consumed.
- ♦ Goal of the analysis of algorithms is to compare algorithms (or solutions).

How to Compare Algorithms:

1) Execution times?

- ♦ Not a Good Idea because execution time will be changed from Machine to Machine or Programming Language to Programming Language

2) Number of statements executed?

- ♦ Not a Good Idea because Number of statements will be changed from Programming Language to Programming Language

What is Best Solution:

Need the Way to Calculate Time Complexity and Space Complexity of an Algorithm without depending on Machine or Programming Language or Developer

7. Rate of Growth Classes

- ♦ Rate at which the running time of algorithm grows as the input size grows is called rate of growth.

Commonly Used Rates of Growth

Complexity	Name
1	Constant
log n	Logarithmic
n	Linear
n log n	Linear Logarithmic
n ²	Quadratic
n ³	Cubic
2 ⁿ	Exponential
3 ⁿ	Exponential
n!	Factorial

Relationship between Rates of Growth Classes:

$$1 < \log n < n < n * \log n < n^2 < n^3 < 2^n < 3^n < n!$$



Ex1: Find the Time Complexity

```
void sum() {  
    int a= 10;  
    int b=20;  
    int sum = a + b;  
    System.out.println(sum);  
}
```

$f(n) = O(1 + 1 + 1 + 1)$
 $= O(4)$
 $= O(1) \Rightarrow$ **Constant Time Complexity**

Ex2: Find the Time Complexity

```
int sum(int n) {  
    System.out.println("Begin");  
    int sum= 0;  
  
    for(int i= 1; i<=n; i++){  
        sum = sum + i;  
    }  
  
    System.out.println("End");  
    return sum;  
}
```

$f(n) = O(1 + 1 + n + n + 1 + 1)$
 $= O(2n + 4)$
 $= O(n) \Rightarrow$ **Linear Time Complexity**

8. Types of Analysis

- ♦ To analyze the given algorithm, We need to know
 - With which inputs the algorithm takes less time
 - With which inputs the algorithm takes a long time.
- ♦ We represent the algorithm with multiple expressions:
- ♦ One for the case where it takes less time and another for the case where it takes more time.
- ♦ There are three types of analysis:
 - a) Best Case
 - b) Average Case
 - c) Worst Case

a) Best Case Analysis

- ♦ Minimum amount of time the Algorithm takes to execute for the given Input.(Faster Execution time)

b) Average Case Analysis

- ♦ Provides a prediction about the running time of the algorithm.
- ♦ Run the algorithm many times, using many different inputs and compute the Average Running Time

c) Worst Case Analysis

- ♦ Maximum amount of time the Algorithm takes to execute for the given Input.(Slower Execution Time)



9. Asymptotic Notations

- ♦ Asymptotic Notations are the mathematical notations used to describe the Time Complexity or Space Complexity of an algorithm
- ♦ There are 3 such Notations:
 - a) Big-O Notation (Upper Bound Function)
 - b) Omega Notation (Lower Bound Function)
 - c) Theta Notation (Tight Bound Function)

10. Big-O Notation

- ♦ Big-O Notation gives tight upper bound of the given function
- ♦ It is Represented as $f(n) = O(g(n))$
i.e Upper bound of $f(n)$ is $g(n)$ at Larger Values of n .

Function $f(n) = O(g(n))$

if there exists +ve constants C and n_0

such that $f(n) \leq C * g(n)$ for all $n \geq n_0$

Ex1:

$$f(n) = 10n + 7$$

$$10n + 7 \leq 11n + 7n \quad \text{for all } n \geq 1$$

$$10n + 7 \leq 18n \quad \text{for all } n \geq 1$$

Here $f(n) = 10n + 7$

$C = 18$

$g(n) = n$

$n_0 = 1$

$n=0$ ----- $7 \leq 0$ // Not Allowed

$n=1$ ----- $17 \leq 18$ // Allowed

$n=2$ ----- $27 \leq 36$ // Allowed

$n=3$ ----- $37 \leq 54$ // Allowed

So $n \geq 1$ is Allowed.

So $f(n) = O(n) \Rightarrow$ Linear Complexity



Ex2:

$$f(n) = 6n^2 + 2n + 32n$$

$$6n^2 + 2n + 32n \leq 6n^2 + 2n^2 + 32n^2 \text{ for all } n \geq 0$$

$$6n^2 + 2n + 32n \leq 40n^2 \text{ for all } n \geq 0$$

$$\text{Here } f(n) = 6n^2 + 2n + 32n$$

$$C = 40$$

$$g(n) = n^2$$

$$n_0 = 0$$

$$n=0 \text{ ----- } 0 \leq 0 \text{ // Allowed}$$

$$n=1 \text{ ----- } 40 \leq 40 \text{ // Allowed}$$

$$n=2 \text{ ----- } 101 \leq 160 \text{ // Allowed}$$

SO $n \geq 0$ is Allowed.

So $f(n) = O(n^2) \Rightarrow$ Quadratic Complexity

Ex3:

$$f(n) = 3n^2 + 5n + 9$$

$$3n^2 + 5n + 9 \leq 4n^2 \text{ for all } n \geq 7$$

$$3n^2 + 5n + 9 \leq 4n^2 \text{ for all } n \geq 7$$

$$\text{Here } f(n) = 3n^2 + 5n + 9$$

$$C = 4$$

$$g(n) = n^2$$

$$n_0 = 7$$

$$n=0 \text{ ----- } 9 \leq 0 \text{ // Not Allowed}$$

$$n=1 \text{ ----- } 17 \leq 4 \text{ // Not Allowed}$$

$$n=2 \text{ ----- } 31 \leq 16 \text{ // Not Allowed}$$

....

$$n=6 \text{ ----- } 147 \leq 144 \text{ // Not Allowed}$$

$$n=7 \text{ ----- } 191 \leq 196 \text{ // Allowed}$$

SO $n \geq 7$ is Allowed.

So $f(n) = O(n^2) \Rightarrow$ Quadratic Complexity



11. Omega Notation

- ♦ Omega Notation gives tight lower bound of the given function
- ♦ It is Represented as $f(n) = \Omega (g(n))$
i.e Lower bound of $f(n)$ is $g(n)$ at Larger Values of n .

Function $f(n) = \Omega (g(n))$
if there exists +ve constants C and n_0
such that $f(n) \geq C * g(n)$ for all $n \geq n_0$

Ex1:

$$f(n) = 3n + 5$$
$$3n + 5 \geq 1n \quad \text{for all } n \geq 0$$
$$3n + 5 \geq 2n \quad \text{for all } n \geq 0$$

Here $f(n) = 3n + 5$

$$C = 2$$

$$g(n) = n$$

$$n_0 = 0$$

$$\begin{array}{ll} n=0 & \text{----- } 5 \geq 0 \quad // \text{ Allowed} \\ n=1 & \text{----- } 8 \geq 2 \quad // \text{ Allowed} \\ n=2 & \text{----- } 11 \geq 4 \quad // \text{ Allowed} \end{array}$$

SO $n \geq 0$ is Allowed.

So $f(n) = \Omega (n) \Rightarrow$ Linear Complexity

Ex2:

$$f(n) = 6n^2 + 2n + 32$$
$$6n^2 + 2n + 32 \geq 5n^2 \quad \text{for all } n \geq 0$$
$$6n^2 + 2n + 32 \geq 5n^2 \quad \text{for all } n \geq 0$$

Here $f(n) = 6n^2 + 2n + 32$

$$C = 5$$

$$g(n) = n^2$$

$$n_0 = 0$$

$$\begin{array}{ll} n=0 & \text{----- } 32 \geq 0 \quad // \text{ Allowed} \\ n=1 & \text{----- } 40 \geq 5 \quad // \text{ Allowed} \end{array}$$

SO $n \geq 0$ is Allowed.

So $f(n) = \Omega (n^2) \Rightarrow$ Quadratic Complexity

Ex3:

$$f(n) = 3n^2 + 5n + 9$$

$$3n^2 + 5n + 9 \geq 2n^2 \quad \text{for all } n \geq 7$$

$$3n^2 + 5n + 9 \geq 2n^2 \quad \text{for all } n \geq 7$$

Here $f(n) = 3n^2 + 5n + 9$

$$C = 2$$

$$g(n) = n^2$$

$$n_0 = 7$$

$$n=0 \quad \text{-----} \quad 9 \geq 0 \quad // \text{ Allowed}$$

$$n=1 \quad \text{-----} \quad 17 \geq 2 \quad // \text{ Allowed}$$

$$n=2 \quad \text{-----} \quad 31 \geq 8 \quad // \text{ Allowed}$$

SO $n \geq 0$ is Allowed.

So $f(n) = \Omega(n^2) \Rightarrow \text{Quadratic Complexity}$

12. Theta Notation

- ♦ Theta Notation gives tight bound of the given function
- ♦ It is Represented as $f(n) = \Theta(g(n))$
i.e Exact bound of $f(n)$ is $g(n)$ at Larger Values of n .
- ♦ Average Running time of an Algorithm is always between lower bound and upper bound.

Function $f(n) = \Theta(g(n))$

if there exists +ve constants c_1, c_2 and n_0

such that $c_1 * g(n) \leq f(n) \leq c_2 * g(n)$ for all $n \geq n_0$



Ex1:

$$f(n) = 5n + 7$$

$$4n \leq 5n + 7 \leq 6n \quad \text{for all } n \geq 7$$

Here $f(n) = 5n + 7$

$$C1 = 4$$

$$C2 = 6$$

$$g(n) = n$$

$$n0 = 7$$

n=0	----	0 ≤ 7 ≤ 0	// Not Allowed
n=1	----	4 ≤ 12 ≤ 6	// Not Allowed
n=2	----	8 ≤ 17 ≤ 12	// Not Allowed
n=3	----	12 ≤ 22 ≤ 18	// Not Allowed
..			
n=6	----	24 ≤ 37 ≤ 36	// Not Allowed
n=7	----	28 ≤ 42 ≤ 42	// Not Allowed

SO $n \geq 7$ is Allowed.

So $f(n) = \Theta(n) \Rightarrow$ Linear Complexity

Ex2:

$$f(n) = 3n^2 + 5n + 9$$

$$4n^2 \leq 3n^2 + 5n + 9 \leq 2n^2 \quad \text{for all } n \geq (\text{You Find})$$

Here $f(n) = 3n^2 + 5n + 9$

$$C1 = 4$$

$$C2 = 2$$

$$g(n) = n^2$$

$$n0 = (\text{You Find})$$

n=0	---	0 ≤ 9 ≤ 0	// Not Allowed
n=1	----	4 ≤ 17 ≤ 2	// Not Allowed
n=2	----	16 ≤ 31 ≤ 8	// Not Allowed
...			

You Calculate n value

So $f(n) = \Theta(n^2) \Rightarrow$ Quadratic Complexity



13. More Examples

Example 1:

```
void sum() {  
    int a= 10;  
    int b=20;  
    int sum = a + b;  
    System.out.println(sum);  
}
```

$f(n) = O(1 + 1 + 1 + 1)$
 $= O(4)$
 $= O(1) \Rightarrow \text{Constant Time Complexity}$

Example 2:

```
int sum(int n) {  
    System.out.println("Begin");  
    int sum= 0;  
  
    for(int i= 1; i<=n; i++){  
        sum = sum + i;  
    }  
  
    System.out.println("End");  
    return sum;  
}
```

$f(n) = O(1 + 1 + n + n + 1 + 1)$
 $= O(2n + 4)$
 $= O(n) \Rightarrow \text{Linear Time Complexity}$



Example 3:

```
void show(int n) {  
    System.out.println("Begin");  
  
    for(int i= 1; i<=n; i++){  
        System.out.println(i);  
    }  
  
    System.out.println("OK");  
  
    for(int i= 1; i<=n; i++){  
        System.out.println(i);  
    }  
  
    System.out.println("End");  
}
```

$$\begin{aligned}f(n) &= O(1 + n + 1 + n + 1) \\&= O(2n + 3) \\&= O(n) \Rightarrow \text{Linear Time Complexity}\end{aligned}$$



Example 4:

```
void show(int n) {  
    System.out.println("Begin");  
  
    System.out.println("OK");  
  
    for(int i= 1; i<=n; i++){  
        for(int j= 1;j<=n; j++){  
            System.out.println(i+"\t"+j);  
        }  
    }  
    System.out.println("OK");  
    System.out.println("End");  
}
```

$$f(n) = O(1 + 1 + (n * n) + 1 + 1)$$

$$= O(n^2 + 4)$$

$$= O(n^2) \Rightarrow \text{Quadratic Time Complexity}$$



Example 5:

```
void show(int n) {  
    System.out.println("Begin");  
  
    System.out.println("OK");  
  
    for(int i= 1; i<=n; i++){  
        System.out.println(i);  
    }  
  
    System.out.println("OK");  
  
    for(int i= 1; i<=n; i++){  
        for(int j= 1; j<=n; j++){  
            System.out.println(i+"\t"+j);  
        }  
    }  
    System.out.println("OK");  
  
    System.out.println("End");  
}
```

$$f(n) = O(1 + 1 + n + 1 + (n * n) + 1 + 1)$$

$$= O(n^2 + n + 5)$$

$$= O(n^2) \Rightarrow \text{Quadratic Time Complexity}$$



Example 6:

```
void show(int n) {  
    System.out.println("Begin");  
  
    System.out.println("OK");  
  
    for(int i= 1; i<=n; i++){  
        System.out.println(i);  
    }  
  
    System.out.println("OK");  
  
    for(int i= 1; i<=n; i++){  
        for(int j= 1;j<=n; j++){  
            for(int k= 1;k<=n; k++){  
  
                System.out.println(i+"\t"+j+"\t"+k);  
            }  
        }  
    }  
  
    System.out.println("OK");  
  
    System.out.println("End");  
}
```

$$\begin{aligned}f(n) &= O(1 + 1 + n + 1 + (n * n * n) + 1 + 1) \\&= O(n^3 + n + 5) \\&= O(n^3) \Rightarrow \text{Cubic Time Complexity}\end{aligned}$$



Example 7:

```
void show(int m,int n) {  
    System.out.println("Begin");  
  
    System.out.println("OK");  
  
    for(int i= 1; i<=n; i++){  
        System.out.println(i);  
    }  
  
    System.out.println("OK");  
  
    for(int i= 1; i<=m; i++){  
        System.out.println(i);  
    }  
  
    System.out.println("OK");  
  
    System.out.println("End");  
}
```

$f(n) = O(1 + 1 + n + 1 + m + 1 + 1)$

$= O(m + n + 5)$

$= O(m+n) \Rightarrow \text{Linear Time Complexity}$



14. Space Complexity

- ♦ Space complexity refers to the total amount of memory space used by an algorithm including the space of input values.
- ♦ Auxiliary space is simply extra or temporary space, and it is not the same as space complexity.

Space Complexity = Auxiliary space + space use by input values

- ♦ The best algorithm/program should have a low level of space complexity.
- ♦ The less space required, the faster it executes.

Example 1:

```
void sum(int a, int b) {  
    int sum = a + b;  
    System.out.println(sum);  
}
```

$f(n) = O(1 + 1 + 1)$

$= O(3)$

$= O(1) \Rightarrow \text{Constant Space Complexity}$



Example 2:

```
int sum(int []arr) {  
    int sum= 0;  
  
    for(int i= 0; i<arr.length; i++){  
        sum = sum + arr[i];  
    }  
    return sum;  
}
```

$f(n) = O(n + 1)$
 $= O(n + 1)$
 $= O(n) \Rightarrow \text{Linear Space Complexity}$

Example 3:

```
void show(int arr[]) {  
  
    int myarr [] = new int[arr.length];  
  
    for(int i= 0; i<n; i++){  
        myarr[i]= arr[i] * arr[i];  
    }  
}
```

$f(n) = O(n + n)$
 $= O(2n)$
 $= O(n) \Rightarrow \text{Linear Space Complexity}$